



Dong-Uk Seo

ROBOTICS ENGINEER, SLAM

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Summary

Ph.D. candidate researching the robotics field in KAIST Urban Robotics Lab. Research interests lie in LiDAR-Camera fusion-based SLAM for robust long-term autonomy. Additionally interested in the process collaborated with multi-agent and localization in the dense map representation.

Education

KAIST(Korea Advanced Institute of Science and Technology)

PH. D. CANDIDATE IN DEPARTMENT OF ELECTRICAL ENGINEERING, URBAN ROBOTICS LAB

- GPA: 4.16 / 4.30

Daejeon, South Korea

Mar. 2023 - Present

KAIST(Korea Advanced Institute of Science and Technology)

M.S. IN DEPARTMENT OF ELECTRICAL ENGINEERING, URBAN ROBOTICS LAB

- GPA: 4.17 / 4.30

Daejeon, South Korea

Mar. 2021 - Feb. 2023

Yonsei University

B.S. IN DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING

- GPA: 4.06 / 4.30
- National Science and Engineering Excellence Scholarship (2016 - 2019, Korea Scholarship Foundation)
- Undergraduate Research Assistant in Compiler Research Laboratory (CORELAB)
- Graduation Thesis: A Deep Learning Based Super-Resolution RF Map Reconstruction Scheme for Fingerprinting Localization

Seoul, South Korea

Mar. 2014 - Aug. 2020

Experiences & Projects

Visual SLAM for indoor autonomous navigation

KAIST URBAN ROBOTICS LAB & SAMSUNG ELECTRONICS

- Visual SLAM is performed indoors using a camera, IMU, and wheel encoder
- Participate as a **team leader**.

Daejeon, South Korea

June. 2024 - Present

Multi-robot collaboration for dense map reconstruction and localization

KAIST URBAN ROBOTICS LAB & YONSEI UNIVERSITY & KOREA UNIVERSITY & NAVER LABS & INCHEON AIRPORT

- While removing dynamic objects, multi-robots collaborate to localize and map with a deep learning-based algorithm for reconstructing a dense representation.
- Participate as a team member.

Daejeon, South Korea

July. 2023 - Present

Robust localization system in a challenging environment for last-mile delivery.

KAIST URBAN ROBOTICS LAB & MOBINN

- An indoor/outdoor day/night localization system for an obstacle-overcoming delivery robot is researched. LiDAR-Visual-Inertial fusion-based SLAM frameworks with deep learning technology are implemented.
- Participate as a **team leader**.

Daejeon, South Korea

May. 2023 - Present

Monocular RGB-D Camera and IMU sensor based Navigation Framework on a mobile robot

KAIST URBAN ROBOTICS LAB & HYUNDAI KEFICO

- Implemented Visual-Inertial SLAM using monocular RGB-D camera and IMU sensor. The global planner and local planner are also implemented on the map refined by Surfel, and finally the robot could proceed with autonomous navigation toward the destination.
- Participated as a **team leader**.

Daejeon, South Korea

April. 2022 - Jan. 2023

Autonomous Driving SLAM framework (Technology transfer)

Daejeon, South Korea

KAIST URBAN ROBOTICS LAB & HILLSLOGIS

Dec. 2021 - Mar. 2023

- Relatively inexpensive sensors such as 2D LiDAR, RGB-D camera, and wheel are used to perform accurate localization and map generation, helping mobile robots to automatically run processes in a factory-like environment. In this project, camera-LiDAR-wheel odometry fusion-based simultaneous localization and mapping (SLAM) was performed.
- **Awarded CES 2023 Innovation Award**
- Participated as a team member.

Visual SLAM on mobile smartphones for AR application

Daejeon, South Korea

KAIST URBAN ROBOTICS LAB & ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE (ETRI)

Mar. 2021 - Present

- For AR application, SLAM is performed on smartphones. To be more general applications, only monocular cameras and IMU sensors have been used, while point and line features are utilized. Also develop an algorithm to recognize planes.
- Participate as a team member.

Last-mile project

Daejeon, South Korea

KAIST URBAN ROBOTICS LAB & KOREA ELECTRONICS TECHNOLOGY INSTITUTE (KETI)

Nov. 2020 - Dec. 2021

- Development of self-driving robot technology capable of learning one's own driving experience and moving to the destination robustly in four seasons, day and night, using various sensor information for driving in indoor and outdoor urban living environments and using uncertain maps.
- Participated as a team member.

Development of big data intelligent information system

Seoul, South Korea

KOREA DATA AGENCY (KDA) & KOREA UNIVERSITY

June. 2019 - Sep. 2019

- Symptom-based Dentist Matching System using BiLSTM-attention.

Publications

* These authors contributed equally

International

Enhancing Robustness of Line Tracking Through Semi-Dense Epipolar Search in Line-based SLAM

Dong-Uk Seo, Hyungtae Lim, EungChang Mason Lee, Hyunjun Lim, Hyun Myung

2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2023

PaGO-LOAM: Robust ground-optimized LiDAR odometry

Dong-Uk Seo*, Hyungtae Lim*, Seungjae Lee, Hyun Myung

2022 19th International Conference on Ubiquitous Robots (UR)

2022

Struct-MDC: Mesh-refined unsupervised depth completion leveraging structural regularities from visual SLAM

Jinwoo Jeon, Hyunjun Lim, **Dong-Uk Seo**, Hyun Myung

IEEE Robotics and Automation Letters (RA-L) 7.3 (2022) pp. 6391–6398. IEEE

2022

QR-SCAN: Traversable Region Scan for Quadruped Robot Exploration using Lightweight Precomputed Trajectory

EungChang Mason Lee, **DongUk Seo**, JinWoo Jeon, Hyun Myung

2021 21st International Conference on Control, Automation and Systems (ICCAS)

2021

Domestic

Surfel-based Depth Image Update for Dynamic Object-Removed Map Generation

Dong-Uk Seo, Sungjae Shin, Hyun Myung

Korea Robotics Society Annual Conference (KRoC 2023)

2023

Optimal Path Planning System using Candidates from Road-Based Sliding Window

Dong-Uk Seo, Hyun Myung

Institute of Control, Robotics and Systems (ICROS) (2022) pp. 421–422

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